

1. Abstract

For image deblurring in real-world scenarios, we propose a two-stage deblurring architecture combining OMDNet and NAFNet, aiming to achieve a complementary effect between the two models. Furthermore, we use visual heatmaps to compare the deblurring quality of each stage.

2. Motivation and Problem Definition

Image blur can be caused by several factors, including incorrect camera focus, atmospheric conditions, camera shake, and object movement. These causes produce different blur patterns, such as defocus blur, atmospheric blur, and motion blur. In this project, we focus on real-world motion blur caused by the relative movement between the camera and objects during exposure. It produces unclear boundaries and unreadable text, especially in traffic and low-light scenes.



Figure 1. Example of real-world motion blur.

Since motion blur may vary across different regions of an image, restoring both overall structures and fine details using a single model remains challenging. The objective of this project is to improve the clarity of real-world motion-blurred images, particularly the overall structures, object boundaries, textures, and text regions.

3. Model & Method

(1) OMDNet

OMDNet focuses on handling non-uniform blur caused by moving objects. Unlike traditional global deblurring, OMDNet can distinguish between dynamic regions and static backgrounds in an image, performing precise deblurring for moving objects, making it suitable as the primary blur removal model in the first stage.

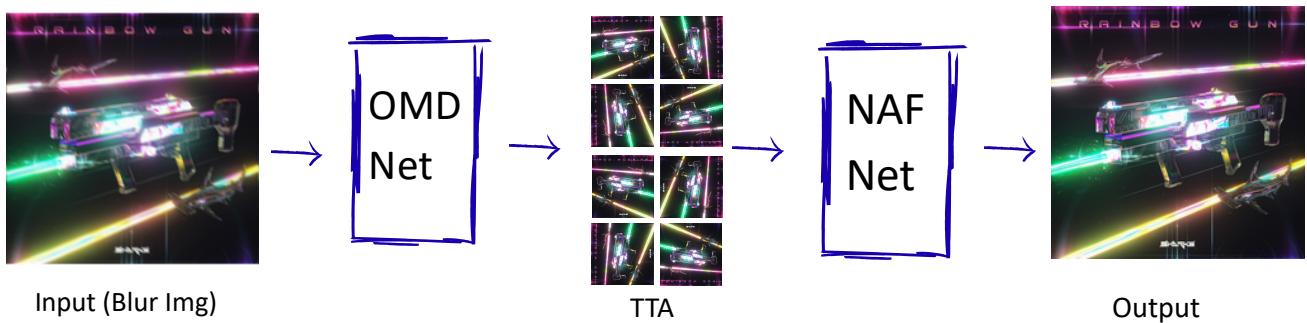
(2) NAFNet

NAFNet's core contribution is abandoning traditional activation functions (ReLU) and improving performance by simplifying the network structure. In deblurring tasks, NAFNet excels at reconstructing detailed textures and sharp edges, making it suitable as a second-stage refinement process.

(3) Test-Time Augmentation(TTA)

To effectively reduce the model's overfitting to specific directions and improve the consistency of edge processing, we introduce the TTA technique, which involves rotating and flipping the input image in eight different ways, then inversely transforming the result back to the original direction and taking the average value.

(4) Pipeline



The input motion-blurred image is first processed by OMDNet to remove large-scale and non-uniform motion blur. The intermediate result is then transformed using eight rotation and flipping combinations for Test-Time Augmentation. Each transformed image is refined by NAFNet, and the outputs are transformed back and averaged to produce the final result. Through this two-stage pipeline, OMDNet restores the overall structure, while NAFNet further improves local edges and textures.

(5) Heatmap

Calculate the pixel difference between the two images and convert it into a grayscale difference map. The greater the difference, the brighter the color of the heat map, which helps us quickly locate the area to be repaired by the model.



4. Experiment Result

(1) NAFNet Only



Input



NAFNet



diff(input, NAF)

(2) OMDNet Only



Input



OMDNet



diff(input, OMD)

(3) OMDNet->NAFNet



OMDNet



OMD+NAF



diff(OMD, OMD+NAF)

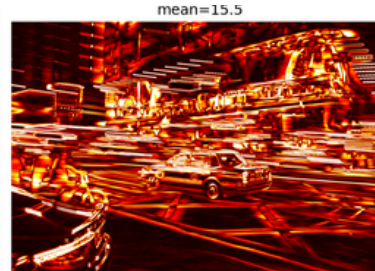
(4) Result



Input



OMD+NAF



diff(input, OMD+NAF)





5. Discussion

Both images we selected are motion-blurred. In the first image, the text on the red ribbon in the upper right corner(頭肩頸)is restored, and the blurred figure below becomes clearer. In the second image, the sign, which was originally completely illegible, is restored(威尼斯), and the text on the taxi also has a clear outline.

From the visual results, the OMDNet->NAFNet pipeline strategy effectively handles motion-blurred scenes, with a clear division of labor between the two models. OMDNet is responsible for large-scale motion blur removal, while NAFNet handles detailed textures, achieving a good deblurring effect.

6. Conclusion

In this project, we proposed a two-stage pipeline combining OMDNet, NAFNet, and Test-Time Augmentation for real-world motion deblurring. OMDNet mainly reduces large-scale and non-uniform motion blur, while NAFNet further refines local textures and object boundaries.

The experimental results demonstrate visible improvements in text readability, edge clarity, and overall image structure. However, severely blurred or completely lost details remain difficult to recover. Future work may include evaluating the method on more diverse images and improving its processing efficiency.

7. Reference

- Chen, L., Chu, X., Zhang, X., & Sun, J. (2022). Simple Baselines for Image Restoration. arXiv Preprint arXiv:2204.04676.
- Wang, Q., Fink, O., Gool, L. V., & Dai, D. (2022). Continual Test-Time Domain Adaptation. <https://arxiv.org/abs/2203.13591>
- Yu, D., Li, J., Zhou, J., Zhuge, Z., Chen, Y., & Li, Q. (2026). OMoBlur: An Object Motion Blur Dataset and Benchmark for Real-World Local Motion Deblurring. Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), 22626–22635.